

Fig. 2: Power supply circuit

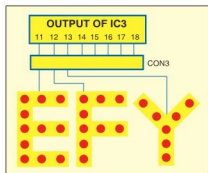


Fig. 3: EFY display using LED strip

interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry.

Ports 1, 2 and 3 of the microcontroller are 8-bit bidirectional I/O ports with internal pull-ups but port 0 is an 8-bit open-drain bidirectional I/O port. Therefore port 0 is pulled up by resistor network RNW1. Port pins P0.0 through P0.7 of the microcontroller are connected to input pins 1 through 8 of IC2, respectively. Similarly, pins of ports 1, 2 and 3 are connected to input pins of IC3 through IC5, respectively, without pull-up resistors.

ULN2803 is a monolithic, high-voltage and high-current Darlington transistor array. It consists of eight npn Darlington pairs that have high-voltage outputs with common-cathode clamp diode for switching inductive loads. The collector-current rating of a single Darlington pair is 500 mA. ULN2803 has a 2.7-kilo-ohm base resistor for each Darlington pair to connect directly with the transistor-transistor logic (TTL) or complementary metal-oxide semiconductor (CMOS) devices.

LED strips are connected to output

pins 11 through 18 of the ULN2803 ICs. The microcontroller drives the LED strips through ULN2803 according to the source program.

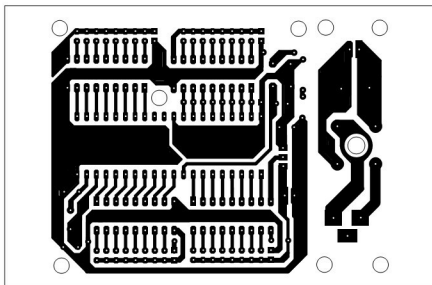


Fig. 4: An actual-size, single-side PCB for microcontroller-based dynamic display using LED strip

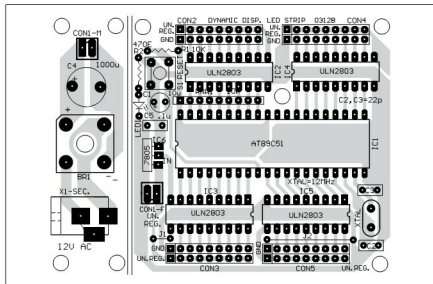


Fig. 5: Component layout for the PCB

Power-on reset for the microcontroller is provided by the combination of resistor R1 and capacitor C1. Switch S1 is used for manual reset. A 12MHz crystal along with two 33pF capacitors connected between pins 18 and 19 provides the basic clock frequency to microcontroller AT89C51.

Fig. 2 shows the circuit for power supply. The 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 5A. The transformer output is rectified by full-wave bridge rectifier BR1, filtered by capacitor C4 and regulated by IC 7805. Capacitor C5 bypasses the